

CO1-AT2

1.
(a) spam classification for email is a supervised machine learning problems.

(i) Naïve Bayes: (selected) $\Rightarrow$ simple, fast, highly effective.

$\Rightarrow$ Based on Bayes's theorem.

$\Rightarrow$ Assumes features are independent.

$\Rightarrow$ For large email datasets.

(ii) Support vector machine (SVM):

$\Rightarrow$ Finds the best boundary separating spam and non-spam emails.

$\Rightarrow$ Gives high accuracy.

$\Rightarrow$ Requires more computation.

(iii) Logistic Regression:

$\Rightarrow$ Predicts the probability that email is a spam.

$\Rightarrow$ Provides good accuracy with balanced dataset.

(b) (i) Tokenization: Split each email into individual words (tokens)

(ii) Stop-word Removal: Removes common words such as: is, the, and, of

(iii) Text cleaning: Convert all text to lowercase.

Removes punctuation, numbers, symbols & extra spaces.

(iv) Stemming: Convert word to their root form. playing $\rightarrow$ play.

(v) vectorization: convert text into numerical values.

TF-IDF (Term Frequency - Inverse Document Frequency)

(c) Evaluation metrics:

Accuracy: % of correctly classified emails.

$$\frac{TP+TN}{TP+TN+FP+FN}$$

Precision: for predicted spam emails are actually spam.

$$\frac{TP}{TP+FP}$$

Recall: measures how many actual spam emails are detected.

$$\frac{TP}{TP+FN}$$

F1-score: Harmonic mean of precision & recall.

$$F1 = \frac{2 \times (\text{precision} \times \text{recall})}{\text{precision} + \text{recall}}$$

2.

(a) Initialization of S and G :

(Specific Boundary) (S)

$\Rightarrow$ Contains the most specific hypothesis.

$\Rightarrow S = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

(General Boundary) (G)

$\Rightarrow$ Contains most general hypothesis

$\Rightarrow G = \langle ?, ?, ?, ? \rangle$

As training example are processed, S becomes more general
more specific.

(b) Example	sky	Airtemp	Humidity	wind	water	Forecast	enjoy sport
1	sunny	warm	normal	strong	warm	same	yes
2	sunny	warm	high	strong	warm	same	yes
3	rainy	cold	high	strong	warm	change	no
4	sunny	warm	high	strong	cool	change	yes

E1: (positive)

$S = \langle \text{sunny, warm, normal, strong, warm, same} \rangle$

$G = \langle ?, ?, ?, ?, ? \rangle$

E2: (positive): Humidity changes

$S = \langle \text{sunny, warm, ?, strong, warm, same} \rangle$

$G =$ remains unchanged.

E3: (negative): general boundary is specialized

G becomes:

$\Rightarrow \langle \text{sunny, ?, ?, ?, ?} \rangle$

$\Rightarrow \langle ?, \text{warm, ?, ?, ?} \rangle$

$\Rightarrow \langle ?, ?, ?, ?, \text{same} \rangle$

(c)

Candidate elimination

Find-S

- => Maintains both S and G boundaries
- => Learns all hypotheses consistent with data.
- => uses both positive & negative example
- => Better generalization capability
- => More Computationally expensive.
- => can represent multiple possible concepts

- => Maintains only the specific hypothesis.
- => learns only one hypothesis.
- => uses only positive example.
- => limited generalization capability
- => simple & faster.
- => cannot represent multiple concepts.

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